AN APPLE WATCH INTERFACE · 2026

Undertone

Join the chorus, a little at a time.

At karaoke, joining a song usually means taking the mic. Undertone lets you bring your voice in one turn of the watch's side wheel at a time, from listening to singing beside the lead, and step back out whenever you like.



DESIGN
Henry Ou

WHAT I DID
Research, interaction and visual design, prototyping, testing

PLATFORM
Apple Watch, with a proposed room audio system

01 THE MOMENT

Karaoke has two volumes: *silent* and *spotlight.*

Someone passes you the song. Take the mic and you're the main act. Stay quiet and you're out of it. Singing along without a mic is a guess: you can't tell how much you stand out, or whether you're drowning out the singer.

TAKE THE MIC	Everyone is listening to you.
SING WITHOUT IT	You have no idea how loud you are.
UNDERTONE	The volumes in between.



Four positions. *One wheel.*

Each turn of the side wheel moves your voice one step further into the song, and each step says what the room now hears. Turn it back down to step out. The screen warms with you.



Listening

MIC OFF

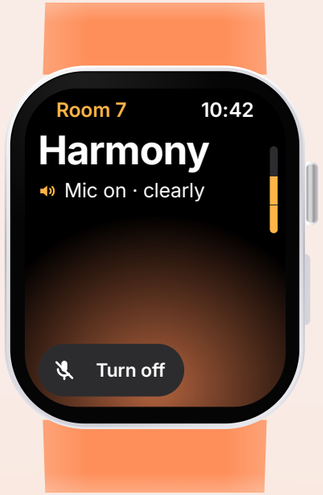
Your voice isn’t amplified. A valid choice, not a failure.



Undertone

≈ -18 dB

Soft, inside the chorus, well under the lead.



Harmony

≈ -8 dB

Clearer, still under the lead.



Duet

≈ 0 dB

Side by side with the lead. Level with her, never above.

↑ ONE TURN UP · ↓ ONE TURN DOWN · ANY TIME

TYPE WEIGHT HERE IS LOUDNESS · dB RELATIVE TO THE LEAD, PROVISIONAL

Only you can turn yourself up.

The room may turn you down, or switch you off. Never up.



ROOM GETS LOUD

Held at Undertone

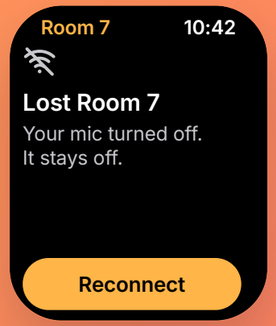
Harmony and Duet drop to Undertone while it lasts. When it clears, you stay put, and may turn up.



SONG IS SOLO

Joining closes

The lead sets it before she sings, and the watch says why. Nobody is turned away in public.



CONNECTION DROPS

Off, and stays off

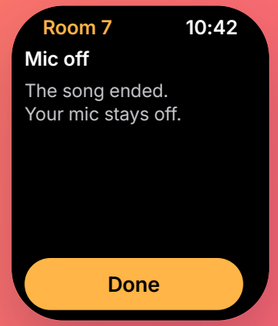
Reconnecting never reopens your mic. You decide when to join again.



WRIST DOWN

A sleeve can’t raise you

Turning up only wakes the screen. Turning down still works.



SONG ENDS

Back to listening

Every song starts with your mic off, as a fresh choice.

04 THE SCREEN

Built for *a glance* in the dark.

A 45 mm screen, a dim room and a song already playing. You should know where you stand without reading anything twice.



1. **Room and time**
   The same top bar Apple's own apps use.
2. **One word for where you are**
   30 pt, bold. Listening counts as a choice.
3. **What the room hears**
   An icon and plain words, so colour is never the only cue.
4. **Three steps beside the wheel**
   One segment per singing level, filling as you turn.
5. **Stage light**
   The screen warms from below as you turn up, so your level reads from the corner of your eye.
6. **Turn off**
   One tap back to listening, always on screen.

You set your level.
The room does the rest.

01 **Your watch**
Controls one thing: your own level.

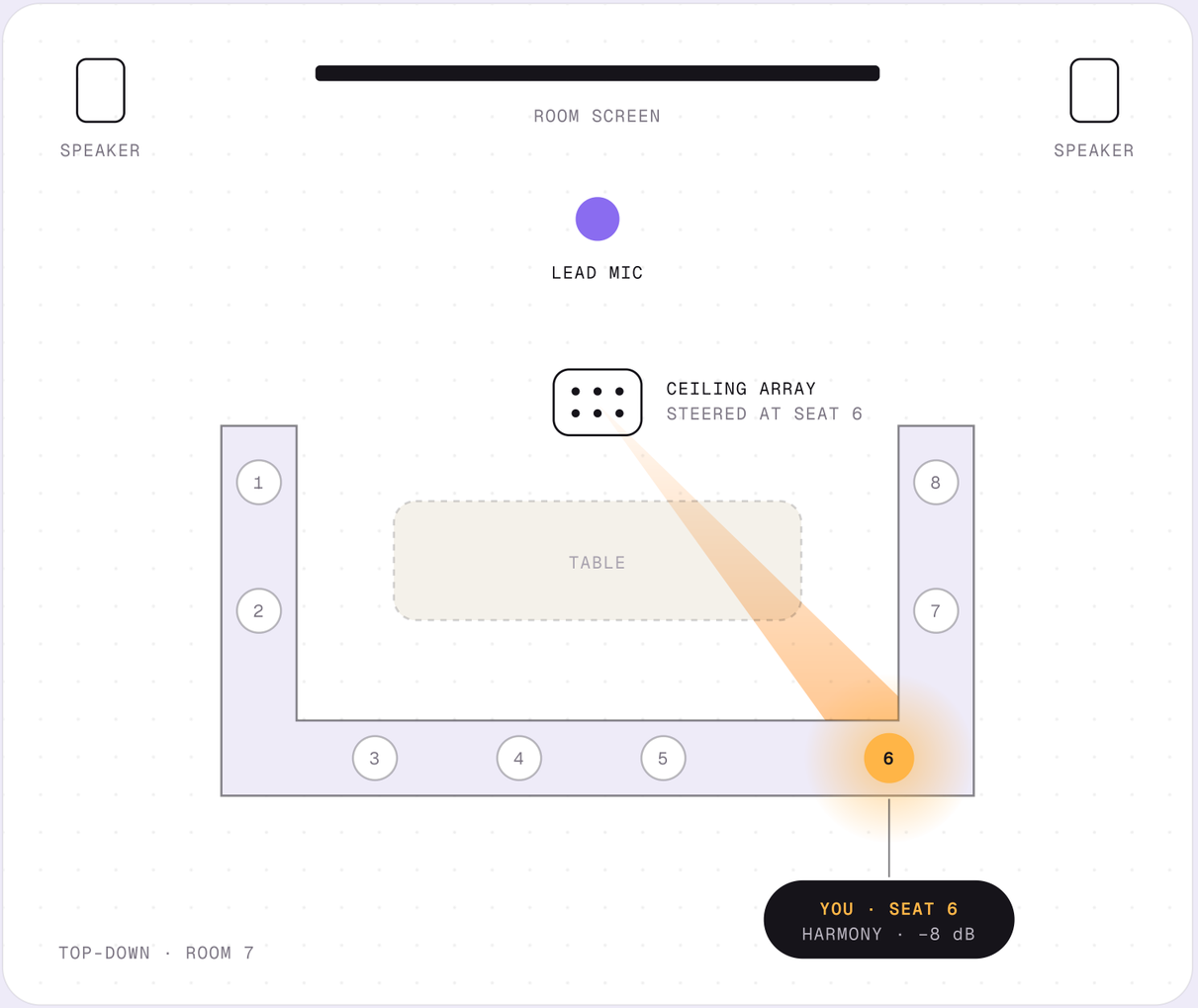
02 **Ceiling microphone array**
Steers a pickup beam at your seat, and only yours.

03 **Room mix**
Places you at your level, never above the lead.

04 **Speakers**
One song, with you somewhere inside it.

A proposal, not a product. The watch interface is the design. The audio system around it is an assumption to test.

SIMULATED IN THE PROTOTYPE · NEIGHBOUR SEPARATION UNTESTED · dB VALUES PROVISIONAL



06 EVERY WAY IN

Works without the wheel.

Honest about the rest.



DESIGNED

Touch version

Join softly, Softer and Louder, with the same rules. Kept separate so the default screen stays quiet.



PARTLY CHECKED

Larger text

An example at about 125% reflows without clipping. 200% isn’t checked yet.



PROPOSED

VoiceOver labels

The level is one adjustable element, so the wheel and a VoiceOver swipe do the same thing.

AGAINST WCAG 2.2 AA, READ FOR A WATCH

● MET	**Text contrast**	Every text colour 7:1 or more; the lowest is 10.1:1
● MET	**Colour is never the only cue**	Every state is written in words; the stage light only supports them
● DESIGNED	**Touch alternative**	Not yet tested with users
● INTENDED	**Touch targets**	40 pt buttons, 44 pt touch areas intended, not measured
● PROPOSED	**Screen reader**	Labels, reading order and announcements for level, noise and connection
● PARTIAL	**Larger text**	About 125% reflows; 200% not checked
● NOT CHECKED	**Keyboard and switch access**	Needs a real watch build

07 LISTENING FIRST

120

public accounts of karaoke nights, gathered in an online ethnography and coded for how people join, hold back or take over a song. Most came from six threads, so they guided decisions rather than proved them.

WHAT THEY HAD IN COMMON

People wanted in, just not all at once, and not at the singer's expense.

E119

"If they have not asked, you can sing along without a microphone."

LED TO A level where you're heard inside the chorus, with no mic in hand

E107

"I sing out of tune and need others to cover my voice a little."

LED TO The first step sits well under the lead

E32 · E37

"I can't hear anyone when EVERYONE is singing!"

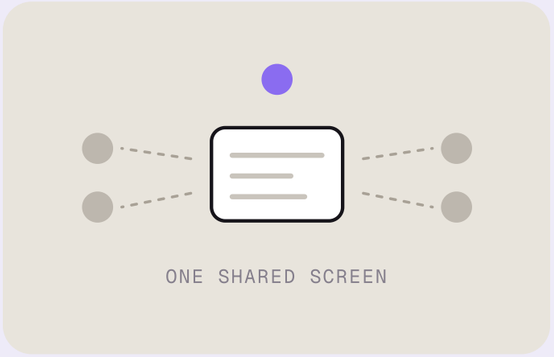
LED TO No one goes above the lead, and room noise holds everyone soft

E112 · E116

Some singers want their song to themselves.

LED TO The lead can keep a song solo before she starts

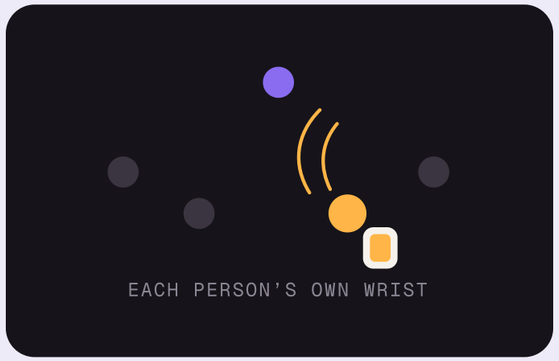
Give the control to the person *deciding* to join.



BEFORE · MY EARLIER CONCEPT

The singer invites, on a shared tablet

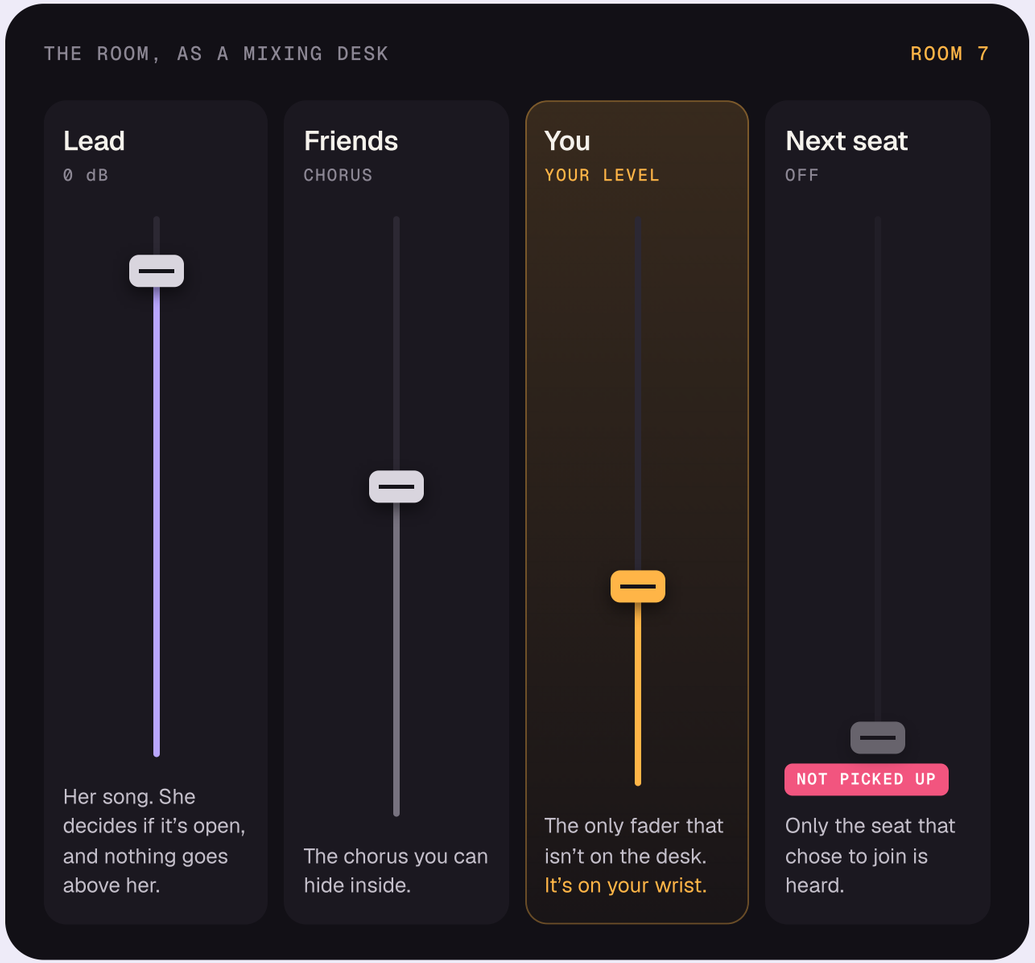
Joining stayed public, and it said little about the person deciding.



AFTER · UNDERTONE

You decide, on your own watch

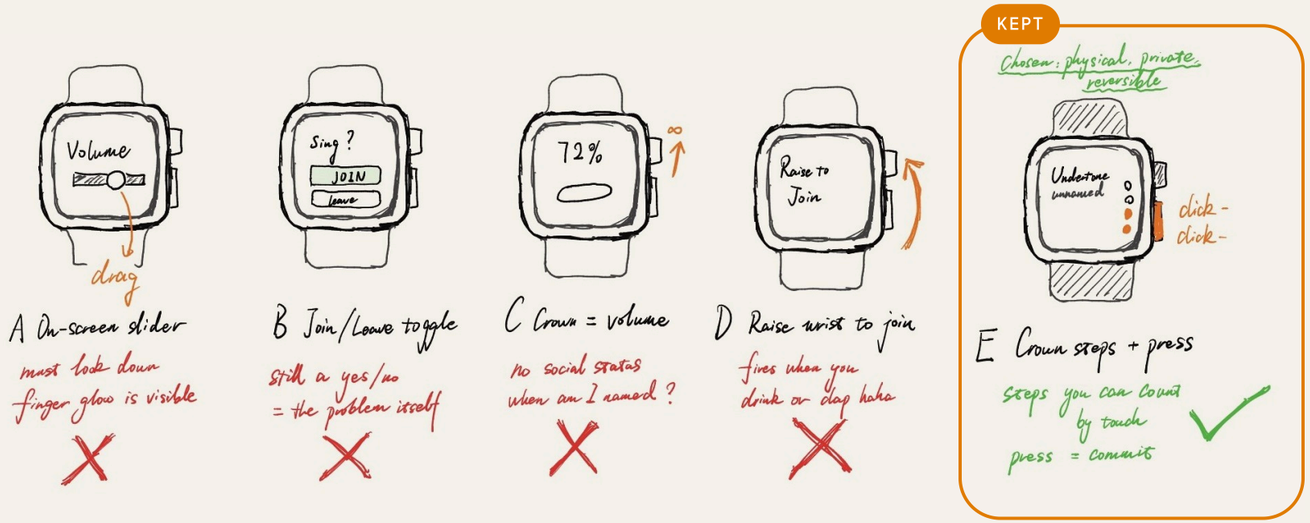
The same problem from the other side. Nobody has to watch you do it.



WHAT TURNED IT AROUND

A critique of the earlier concept: it ignored the person deciding whether to join.

Five decisions, *sketched before any screen.*



SHEET 1 · INPUT

Five ways to control joining, drawn by hand. Stepped turns of the side wheel won: turning maps to “more or less of me”, the steps can be counted by feel, and nobody sees you do it.

1 **Input** — KEPT: WHEEL STEPS

A slider needs a look; a join toggle is all or nothing.

2 **Layout** — KEPT: ONE WORD

A ring read as a volume meter, a list as a menu.

3 **Visibility** — KEPT: PRIVATE

Who is singing is never shown or counted, on any screen.

4 **Naming** — CUT

Naming the person who joins added risk and no benefit.

5 **Leaving** — CHANGED

The palm cover I ticked can’t be promised on watchOS. Turning down became the exit, plus a button.

10 THE CUTTING ROOM

Less, until it did *three things.*

~~Your name on the room screen~~	A second goal, and a turn too far could name you by accident.
~~A one-time tip screen~~	The guidance moved onto Listening itself.
~~A score after the song~~	It judges the very fear the design works against.
~~Volume that follows your voice~~	Takes control away from you.
~~The singer invites you by name~~	Puts you on the spot it was meant to avoid.

Join. Adjust. Stop.

HOW THE FLOW CHANGED

v1.0
A seat picker, three levels, and a named duet behind a side-button press.

v1.1
watchOS reserves that press, so it became a tap. Wrist down can’t turn you up.
SELF-CRITIQUE · AI HEURISTIC REVIEW

v1.2
The lead can keep a song solo. Turning down becomes the main way out.
DESIGN REVIEWS

v1.3
Rebuilt on native watch patterns: one word, a level bar, a bottom button.
CHECKING AT TRUE SIZE

Tested
Duet, naming and the tip removed. Listening, Undertone and Harmony.
PEER REVIEW

Now
Clearer labels from testing. The duet returns as the top step, without a name.
AFTER TESTING · NOT YET TESTED

Three think-aloud sessions. *People got the idea, and tripped on words.*

3/3

could explain why joining a solo song was closed

2/3

joined without help; the third needed a hint to find the wheel

1

mouse scroll skipped a level, and was corrected by the participant

2/3

stopped on their own; the third tapped the mic button, result not recorded

WHO · ON A LAPTOP, A MOUSE WHEEL AS THE SIDE WHEEL

P1	Rarely sings · 5–6 karaoke visits · no Apple Watch
P2	Likes singing · goes often · wears a Watch daily
P3	Mostly listens · two visits · had seen screenshots

International students, two first-year and one second-year, all friends of mine, so their comments may lean kind.

TASK	P1	P2	P3
Enter and confirm the seat	●	●	●
Join very softly	●	●	◐
Be heard more clearly	●	●	●
Try to join a solo song	●	●	●
Go silent after a break	●	●	○

● Alone ◐ After a hint ○ Not recorded

HEARD IN THE ROOM

“What does Turn the crown mean?”

P3 · FINDING THE CONTROL

“Am I too loud?”

P1 · READING THE NOISE WARNING

“She wants to sing this one on her own.”

P1 · THE SOLO RULE

Four words, *then one more step.*

“Does Join quietly turn the mic on?”

~~Join quietly~~ **Enter room**

P3 · with “Mic stays off” said first

“Which dot is me?”

~~Your seat~~ **You: seat 6**

P1 · P2 tried tapping a seat

“What does Turn the crown mean?”

~~Turn the crown to join~~ **Turn wheel to join**

P3 · with how to stop on the same screen

“Am I too loud?”

~~Too loud near you~~ **Room noise is high**

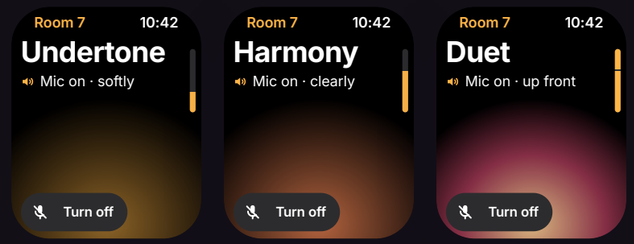
P1 · names the cause, not the person

Quotes translated from Chinese. In a follow-up, P3, who had seen the earlier version, found the revised flow clearer; not a first impression, so a retest with someone new is still needed.

AFTER TESTING · NOT YET TESTED

The scale now ends *beside the lead.*

In testing, one participant reached Harmony and tried one more turn. The level bar already had a third segment that only the cut duet ever used. So the duet came back as the top step, without a name, reached one turn at a time like every other level, and the screen now warms as you rise.



The wheel steps stayed as they were: one mouse scroll skipped a level, but a mouse says little about a real watch wheel.

13 OUTRO

What I'd keep, *and what's still open.*

WHY A WATCH

Already on your wrist

A phone has to be unlocked and held up; a shared tablet makes joining public. The cost is a tiny screen with no room to explain, which is why every word mattered.

THE DECISION THAT MATTERED

Control moved to the joiner

Moving it from the singer's tablet to the joiner's wrist changed who the design serves. One rule then shaped every other screen.

WHAT CRITIQUE CHANGED

I stopped adding

Extra features competed with joining. Testing showed people understood the levels and stumbled on words, so I fixed the words.

A TRADE-OFF I KEPT

The native look

I expect an Apple-like screen to feel trustworthy on a watch, though that's untested. The touch version stays separate so the default screen stays quiet.

STILL OPEN

- HOW LOUD EACH LEVEL SOUNDS TO OTHERS
- REAL WHEEL STEPS ON A WATCH
- WHETHER FRIENDS RECOGNISE A VOICE
- SEPARATING NEIGHBOURING VOICES
- DUET AND THE STAGE LIGHT WITH USERS
- WHETHER DUET SHOULD NEED THE LEAD'S OK
- VOICEOVER ON A REAL WATCH

NEXT STUDY

A real watch, in a *real karaoke room.*

The desktop test shows the interface is understood, not that joining feels safer. Next: friend groups, a watch build, and a person at the mixing desk standing in for the audio system.

01 Felt exposure after each song, 1–7

02 Glances per level change

03 Time to stop mid-chorus

04 Whether friends can tell who sang along

05 How lead singers rate songs with and without joiners

14 PLAY IT

Now turn the wheel.

The prototype runs in your browser. Join softly, go all the way to Duet, make the room loud, ask for a solo, and hear your own voice come in under the lead.

yulinhenryou.github.io/undertone

Scroll the watch, drag the side wheel, or press ↑ and ↓. Sound on.



CREDITS

Research, interaction and visual design, prototyping and testing by Henry Ou. Test participants are anonymised as P1–P3.

TOOLS AND AI

The design is mine. Claude helped with critique, editing, and building this site, the film and the prototype to my direction; Codex gave design reviews. The photographic karaoke scene is AI-generated.

TYPE

Geist, Geist Mono and Instrument Serif. Inter on the watch, standing in for SF Compact. All under the SIL Open Font License.

NOTES

Device frames are drawn in SVG, not Apple product images. Apple Watch is a trademark of Apple Inc. The song in the prototype is original.